

Chapter 7, Problem 82P

Problem

In designing a signal-switching circuit, it was found that a $100\text{-}\mu\text{F}$ capacitor was needed for a time constant of 3 ms. What value resistor is necessary for the circuit?

Step-by-step solution

Step 1 of 1

$$\tau = RC \rightarrow (1)$$

$$R = \frac{T}{C} = \frac{3 \times 10^{-3}}{100 \times 10^{-6}} = 30 \Omega .$$